

Non-Normality, Heteroscedasticity and Serial Correlation

The exact theory of Part III rests on three assumptions about the errors: they are normal, they have a common variance, and they are uncorrelated. [Chapter 19](#) worked out what least squares does when each of them fails. This chapter is about the data analyst's side of the same question: how to see a failure in the residuals, how large it must be before it matters, and what to do about it. The three assumptions do not matter equally. Coefficient inference survives nonnormal errors in moderately large samples, but prediction intervals do not. Unequal variances leave least squares unbiased but make its usual standard errors wrong, sometimes by a factor of four or more, and they can be repaired either by weighting or by a variance estimator that does not trust the model. Serial correlation does the same damage in time-ordered data, and it often signals a mean function that is missing its dynamics. Every diagnostic in the chapter is computed from residuals, which are not the errors. How they differ is where the chapter starts.

What you need. Residuals, leverage and generalized least squares from [Chapter 6](#) (Proposition 6.4.3, Proposition 6.8.1, Definition 6.9.1, Theorem 6.9.1), the Gauss–Markov and Aitken theorems and the normal sampling theory of [Chapter 7](#) (Theorem 7.2.1, Corollary 7.2.3, Theorem 7.5.1), the moments of quadratic forms from [Chapter 2](#) (Theorem 2.3.1, Theorem 2.3.3) and their normal distribution theory from [Chapter 4](#) (Theorem 4.2.1, Corollary 4.3.3), the deletion formulas of [Chapter 10](#) (Proposition 10.6.2), and prediction intervals from [Chapter 12](#) (Theorem 12.3.1) and their large-sample coverage under nonnormal errors (Proposition 12.3.3). From [Chapter 19](#) we use the covariance of least squares under a general error covariance (Theorem 19.2.2) and its large-sample behaviour under nonnormal errors (Theorem 19.3.2). From [Chapter 20](#) we use studentized residuals (Definition 20.1.1, Theorem 20.3.3). [Section 21.3](#) uses the moments of least squares and of s^2 from [Chapter 5](#) (Theorem 5.5.1, Theorem 5.6.1), and [Section 21.5](#) the extremal Rayleigh quotient of [Chapter 1](#) (Theorem 1.7.7) together with the behaviour of a normal vector under a linear map (Theorem 3.3.1, [Chapter 3](#)). The limit theorems of probability that the chapter needs are gathered, or pointed to, in [Section 21.2](#).

21.-1.1 Roadmap

- [Checking normality](#)
- [Detecting heteroscedasticity](#)
- [Weighted least squares](#)
- [Heteroscedasticity-consistent covariance estimators](#)
- [Serial correlation and the Durbin–Watson test](#)
- [Autoregressive errors](#)
- [Summary and notes](#)